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Vehicle cowl structure

Abstract

A vehicle cowl structure includes a cowl body, a cowl panel, an air outlet, and a chamber. The cowl body has a recessed shape with an upper opening facing a vehicle upper side when viewed in a longitudinal direction thereof. The cowl panel closes the upper opening of the cowl body and has an outside air inlet. The outside air inlet allows outside air to be introduced into the cowl body. The air outlet is disposed in the cowl body and allows air inside the cowl body to be discharged toward a vehicle interior. The chamber has a hollow interior, is provided in a bottom part of the cowl body, and is disposed between the outside air inlet and the air outlet in a vehicle width direction. The chamber includes a blocking wall, a chamber-side drainage outlet, and a communication port. The blocking wall has a water inlet.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) The present application claims priority from Japanese Patent Application No. 2022-044776 filed on Mar. 21, 2022, the entire contents of which are hereby incorporated by reference.
BACKGROUND
(2) The disclosure relates to a vehicle cowl structure.
(3) A vehicle cowl structure is configured to, when outside air is introduced into a cowl, allow air and liquid such as rainwater contained in the outside air to be separated from each other, and to send the separated air to an air conditioner side. For example, Japanese Unexamined Patent Application Publication (JP-A) No. 2007-125995 discloses a cowl structure in which a cowl duct is provided in a cowl and partitions the inside of the cowl. The cowl duct has a hole through which air is sent to the air conditioner side and a rib around the hole. Thus, when outside air is introduced into the cowl, the rib of the cowl duct allows air and liquid such as rainwater contained in the outside air to be separated from each other.
SUMMARY
(4) An aspect of the disclosure provides a vehicle cowl structure. The vehicle cowl structure includes a cowl body, a cowl panel, an air outlet, and a chamber. The cowl body extends in a vehicle width direction on a vehicle front side of a lower end of a windshield glass of the vehicle and has a recessed shape with an upper opening facing a vehicle upper side when viewed in a longitudinal direction of the cowl body. The cowl panel extends in the vehicle width direction on the vehicle upper side of the cowl body, closes the upper opening of the cowl body, and has an outside air inlet. The outside air inlet allows outside air to be introduced into the cowl body. The air outlet is disposed in the cowl body and allows air inside the cowl body to be discharged toward a vehicle interior of the vehicle. The chamber has a hollow interior serving as a chamber interior, is provided in a bottom part of the cowl body, and is disposed between the outside air inlet and the air outlet in the vehicle width direction. The chamber includes a blocking wall, a chamber-side drainage outlet, and a communication port. The blocking wall constitutes a wall of the chamber on a side of the outside air inlet, extends upward from a bottom wall of the cowl body, and has, at a lower end of the blocking wall, a water inlet. The water inlet allows liquid inside the cowl body to be drawn into the chamber interior. The chamber-side drainage outlet allows the liquid drawn into the chamber interior to be drained out. The communication port is disposed on a side of the air outlet side with respect to the blocking wall and allows the chamber interior to communicate with an inside of the cowl body.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The accompanying drawings are included to provide a further understanding of the disclosure and are incorporated in and constitute a part of this specification. The drawings illustrate an embodiment and, together with the specification, serve to describe the principles of the disclosure.
(2) FIG. **1** is a partially schematic, top plan view of a front part of a vehicle to which a vehicle cowl structure according to the present embodiment is applied;
(3) FIG. **2** is a schematic, front sectional view (enlarged sectional view taken along line **2-2** in FIG. **1**) of an inside of a left part of the vehicle cowl structure illustrated in FIG. **1**;
(4) FIG. **3** is a schematic, right sectional view (enlarged sectional view taken along line **3-3** in FIG. **2**) of an inside of the vehicle cowl structure illustrated in FIG. **2**;
(5) FIG. **4** is a top plan view of a chamber illustrated in FIG. **2**; and
(6) FIG. **5** is a schematic, front view illustrating a modification of the vehicle cowl structure according to the present embodiment.
DETAILED DESCRIPTION

(7) Recent trend is that a passageway for air in a cowl is relatively narrow when a wiper unit, an airbag, and the like are mounted in the cowl. This results in a relatively high flow velocity of outside air flowing through the cowl, and difficulty in achievement of satisfactory separation between air and liquid contained in the outside air. Thus, the cowl structure is desired to achieve satisfactory separation between the liquid and the air.
(8) It is desirable to provide a vehicle cowl structure that makes it possible to achieve satisfactory separation between liquid and air.
(9) In the following, an embodiment of the disclosure is described in detail with reference to the accompanying drawings. Note that the following description is directed to an illustrative example of the disclosure and not to be construed as limiting to the disclosure. Factors including, without limitation, numerical values, shapes, materials, components, positions of the components, and how the components are coupled to each other are illustrative only and not to be construed as limiting to the disclosure. Further, elements in the following example embodiment which are not recited in a most-generic independent claim of the disclosure are optional and may be provided on an as-needed basis. The drawings are schematic and are not intended to be drawn to scale. Throughout the present specification and the drawings, elements having substantially the same function and configuration are denoted with the same numerals to avoid any redundant description.
(10) A vehicle cowl structure **10** (hereinafter, simply referred to as the cowl structure **10**) according to the present embodiment will be described with reference to the drawings. Note that, arrow UP, arrow FR, and arrow RH, which are illustrated as appropriate in the drawings, indicate a vehicle upper side, a vehicle front side, and a vehicle right side (a first side in a vehicle width direction), respectively, of a vehicle (automobile) V to which the cowl structure **10** is applied. In the following description, unless otherwise specified, an up-down direction, a front-rear direction, and a left-right direction refer to a vehicle up-down direction, a vehicle front-rear direction, and a vehicle left-right direction, respectively.
(11) Overall Configuration
(12) As illustrated in FIG. **1**, the cowl structure **10** is disposed on a front side of a lower end (front end) of a windshield glass **40** of the vehicle V, and on a lower side of a rear end of a hood **42** of the vehicle V. The cowl structure **10** has a substantially cylindrical shape extending in the vehicle width direction as a whole. An inside of the cowl structure **10** serves as a cowl duct part **16** (see FIG. **3**). The cowl structure **10** has outside air inlets **14**C and an air outlet **12**H (see FIG. **2**). Each of the outside air inlets **14**C is configured to allow outside air (air) to be introduced into the cowl duct part **16**. The air outlet **12**H is configured to allow the air introduced into the cowl duct part **16** to be discharged toward a vehicle interior. One end of an air-conditioning duct **44** of the vehicle V is coupled to the air outlet **12**H. The air discharged from the air outlet **12**H into the air-conditioning duct **44** is supplied into the vehicle interior through air vents **44**A. The air vents **44**A constitute the other ends of the air-conditioning duct **44**. An air conditioner **46** is provided to an intermediate part of the air-conditioning duct **44** and includes a blower fan **46**A. The blower fan **46**A is operated to generate, in the cowl duct part **16**, airflow directed from the outside air inlets **14**C to the air outlet **12**H.
(13) Cowl Structure **10**
(14) As illustrated in FIGS. **1** to **4**, the cowl structure **10** includes a cowl body **12**, a cowl top panel **14**, and a gas-liquid separation mechanism **20**. In one embodiment, the cowl top panel **14** may serve as a “cowl panel”.
(15) Cowl Body **12**
(16) The cowl body **12** is made of a metal plate. The cowl body **12** extends in the vehicle width direction and has longitudinal ends joined to an apron member (not illustrated). The apron member constitutes a frame member of the vehicle V. The cowl body **12** has a recessed shape with an upper opening when viewed in the longitudinal direction thereof. In one example, the cowl body **12** includes a bottom wall **12**A, a front wall **12**B, and a rear wall **12**C. The front wall **12**B extends

upward from a front end of the bottom wall **12**A. The rear wall **12**C extends upward from a rear end of the bottom wall **12**A.

(17) A flange **12**D bent forward is formed at an upper end of the front wall **12**B of the cowl body **12**. A flange **12**E is formed at an upper end of the rear wall **12**C of the cowl body **12**. The flange **12**E is bent rearward and obliquely upward and disposed below the lower end (front end) of the windshield glass **40**. The rear wall **12**C of the cowl body **12** is coupled to a dash panel (not illustrated) that partitions an engine room and the vehicle interior of the vehicle V. The rear wall **12**C of the cowl body **12** has, at its left end, the air outlet **12**H (see FIG. **2**) penetrating therethrough. The air outlet **12**H has a substantially rectangular shape whose longitudinal direction corresponds to the left-right direction.

(18) Cowl Top Panel **14**

(19) The cowl top panel **14** is made of a resin material. The cowl top panel **14** has a substantially long plate shape extending in the vehicle width direction. A flange **14**A is formed at a front end of the cowl top panel **14**. The flange **14**A protrudes forward, is disposed on the flange **12**D of the cowl body **12**, and is fixed to the flange **12**D by a fastening member such as a clip. A panel coupler **14**B (see FIG. **3**) is formed at a rear end of the cowl top panel **14**. The panel coupler **14**B has a substantially U shape that is open rearward and obliquely upward when viewed in the longitudinal direction of the cowl top panel **14**. The lower end of the windshield glass **40** is inserted into the panel coupler **14**B. The panel coupler **14**B is fixed to the flange **12**E of the cowl body **12** by a fastening member such as a clip. Thus, the upper opening of the cowl body **12** is closed by the cowl top panel **14**. The inside of the cowl structure **10** defined by the cowl body **12** and the cowl top panel **14** serves as the cowl duct part **16**.

(20) The cowl top panel **14** has, at its right end, outside air inlets **14**C penetrating therethrough. The outside air inlets **14**C each have a substantially long hole shape whose longitudinal direction corresponds to a substantially front-rear direction, and are disposed side by side at predetermined intervals in the left-right direction. This allows communication, through the outside air inlets **14**C, between the inside and the outside of the cowl duct part **16**.

(21) Gas-Liquid Separation Mechanism **20**

(22) As illustrated in FIGS. **2** to **4**, the gas-liquid separation mechanism **20** is provided in the bottom part of the cowl duct part **16**, and is disposed between the outside air inlets **14**C and the air outlet **12**H. In one example, the gas-liquid separation mechanism **20** is provided in the bottom part of the left part of the cowl body **12** and is disposed on the right side of the air outlet **12**H. The gas-liquid separation mechanism **20** is configured to, when outside air (air) is introduced into the cowl duct part **16**, allow air and liquid such as rainwater contained in the outside air to be separated from each other.

(23) The gas-liquid separation mechanism **20** includes, as a main part, a chamber **22** having a hollow interior. The chamber **22** is provided on the bottom wall **12**A of the cowl body **12** and is disposed on the right side of the air outlet **12**H. The chamber **22** has a substantially hollow wedge shape protruding rightward and obliquely upward from the bottom wall **12**A of the cowl body **12** when viewed from the front side. In one example, the chamber **22** includes a blocking wall **24** and a partition wall **26**. The blocking wall **24** constitutes a right wall of the chamber **22**. The partition wall **26** extends leftward from an upper end of the blocking wall **24**. The blocking wall **24** and the partition wall **26** constitute an outer shell of the chamber **22**.

(24) The blocking wall **24** extends in the up-down direction and is inclined rightward (outside air inlet **14**C side) toward an upper side when viewed from the front side. The lower end of the blocking wall **24** is bent rightward and joined to the bottom wall **12**A of the cowl body **12**. An upper end of the blocking wall **24** is disposed in an up-down intermediate part of the cowl duct part **16** and is disposed at a position spaced apart downward from the cowl top panel **14**.

(25) The partition wall **26** extends leftward from the upper end of the blocking wall **24**. The partition wall **26** is smoothly inclined downward in a curved manner toward the left side when

viewed from the front side. A left end (lower end) of the partition wall **26** is disposed to be in contact with the bottom wall **12**A of the cowl body **12** and is joined to the bottom wall **12**A. The chamber **22** is disposed adjacent to and between the front wall **12**B and the rear wall **12**C of the cowl body **12** (see FIG. **3**). That is, the chamber **22** is provided over the entire length of the cowl duct part **16** in the front-rear direction. Consequently, the interior of the chamber **22** is defined by the blocking wall **24**, the partition wall **26**, and the bottom wall **12**A, the front wall **12**B, and the rear wall **12**C of the cowl body **12**. The defined interior space serves as a chamber interior **28**.
(26) Since the chamber **22** is disposed at the lower part (bottom part) of the cowl duct part **16**, a space above the chamber **22** in the cowl duct part **16** is narrowed by the chamber **22**. The space above the chamber **22** in the cowl duct part **16** serves as a duct throttle part **30**. A cross-sectional area of the duct throttle part **30** is set to be smaller than a cross-sectional area of another part in the cowl duct part **16** and set to be larger toward the left side. Furthermore, an up-down dimension H**1** of the blocking wall **24** is set to be larger than an up-down dimension H**2** of an inlet **30**A which is a right end of the duct throttle part **30** (see FIG. **3**).
(27) The blocking wall **24** has, at its lower end, multiple (in the present embodiment, five) water inlets **24**A penetrating therethrough (see FIG. **3**). The water inlets **24**A each have a long hole shape whose longitudinal direction corresponds to the up-down direction, and are disposed side by side in the front-rear direction. This allows communication, through the water inlets **24**A, between (the right space of) the cowl duct part **16** and the chamber interior **28**, and the liquid flowing leftward in the cowl duct part **16** to be drawn into the chamber interior **28** through the water inlets **24**A. The bottom wall **12**A of the cowl body **12** constituting the chamber interior **28** has multiple (in the present embodiment, three) chamber-side drainage outlets **32** penetrating therethrough (see FIGS. **2** and **4**). The chamber-side drainage outlets **32** are disposed side by side in the front-rear direction. Then, the liquid flowing into the chamber interior **28** is drained out of the cowl duct part **16** through the chamber-side drainage outlets **32**. The partition wall **26** has multiple (in the present embodiment, three) communication ports **26**A penetrating therethrough (see FIGS. **2** and **4**). The communication ports **26**A are disposed side by side in the front-rear direction. This allows communication, through the communication ports **26**A, between the chamber interior **28** and the duct throttle part **30**.
(28) The bottom wall **12**A of the cowl body **12** has, between the chamber **22** and the air outlet **12**H, multiple (in the present embodiment, three) body-side drainage outlets **34** penetrating therethrough (see FIGS. **2** and **4**). The body-side drainage outlets **34** are disposed side by side in the front-rear direction. The liquid having passed through the duct throttle part **30** is drained out of the cowl duct part **16** from the body-side drainage outlets **34**.
(29) Furthermore, the gas-liquid separation mechanism **20** has an inclined wall **36**. The gas-liquid separation mechanism **20** is disposed on the left side of the chamber **22** and the body-side drainage outlets **34** and on the right side of the air outlet **12**H, and is disposed above the bottom wall **12**A of the cowl body **12**. The inclined wall **36** is inclined upward toward the left side when viewed from the front side, and is disposed below the air outlet **12**H. In the present embodiment, the inclined wall **36** and the chamber **22** are formed in one piece. The body-side drainage outlets **34** penetrate, in the up-down direction, a part coupling the front end of the inclined wall **36** to the rear end of the chamber **22**.
(30) Operation and Effects
(31) Operation and effects of the present embodiment will now be described.
(32) In the cowl structure **10** with the above structure, the outside air inlets **14**C are formed at the right end of the cowl top panel **14**, and the air outlet **12**H is formed at the left end of the cowl body **12**. When the air conditioner **46** of the vehicle V is turned on, the blower fan **46**A of the air conditioner **46** is operated to allow the air in the cowl duct part **16** to be drawn in by suction from the air outlet **12**H toward the air-conditioning duct **44**. Consequently, the outside air (air) is introduced into the cowl duct part **16** through the outside air inlets **14**C, and airflow directed from

the outside air inlets **14**C to the air outlet **12**H is generated in the cowl duct part **16**.
(33) Here, water (liquid) such as rainwater may enter the cowl duct part **16** through the outside air inlets **14**C together with air. At this time, airflow AR**1** containing liquid water having relatively large particles flows leftward along the bottom wall **12**A side of the cowl body **12** in the lower part of the cowl duct part **16**. Consequently, as illustrated in FIG. **2**, the airflow AR**1** flowing along the bottom wall **12**A side flows into the chamber interior **28** through the water inlets **24**A of the blocking wall **24** of the chamber **22**. Then, the water flowing into the chamber interior **28** is drained out of the cowl duct part **16** from the chamber-side drainage outlets **32** (see arrow B in FIG. **2**), and the air flowing into the chamber interior **28** flows out from the communication ports **26**A to the duct throttle part **30** side (see arrow A in FIG. **2**). That is, the chamber **22** allows the air and the water having the relatively large particles contained in the airflow AR**1** to be separated from each other, the separated water to be drained from the chamber-side drainage outlets **32**, and the separated air to flow from the communication ports **26**A toward the air outlet **12**H. For example, when a large amount of water flows into the cowl duct part **16**, the chamber **22** can achieve effective separation between water and air.
(34) For example, when water droplets are scattered on the inside of the cowl duct part **16**, airflow AR**2** containing water droplets flows leftward on a substantially center side of the cowl duct part **16** and hits the blocking wall **24** of the chamber **22**. This reduces the flow velocity of the airflow AR**2** hitting the blocking wall **24**. In addition, the blocking wall **24** is inclined rightward (upstream side of the airflow AR**2**) toward the upper side when viewed from the front side. In other words, the inclined wall **36** is inclined upward (downstream side of the airflow AR**2**) toward the left side when viewed from the front side. Thus, the airflow AR**2** hitting the blocking wall **24** creates a vortex flow directed downward along the blocking wall **24** (see the airflow AR**2** illustrated in FIG. **2**). As a result, the vortex flow allows air and water contained in the airflow AR**2** to be separated from each other.
(35) For example, the airflow AR**2** causes the separated water to flow downward along the blocking wall **24**. Thus, in the same manner as described above, the water flows into the chamber interior **28** from the water inlets **24**A of the blocking wall **24**, and the water flowing into the chamber interior **28** is drained out of the cowl duct part **16** from the chamber-side drainage outlets **32**.
(36) Furthermore, for example, atomized water having relatively small particles contained in the separated water is drawn into the duct throttle part **30** by airflow AR**3** (see FIG. **2**) flowing leftward through the duct throttle part **30** of the cowl duct part **16**. The atomized water drawn into the duct throttle part **30** flows together with the airflow AR**3** flowing leftward along the bottom surface of the duct throttle part **30** (that is, the upper surface of the partition wall **26**). Consequently, the water contained in the airflow AR**3** adheres to the bottom surface of the duct throttle part **30** (that is, the upper surface of the partition wall **26**) and flows to the bottom wall **12**A side of the cowl body **12** along the partition wall **26**. Then, the water flowing along the partition wall **26** is drained out of the cowl duct part **16** from the body-side drainage outlets **34** (see arrow C in FIG. **2**).
(37) As described above, when the airflow AR**3** passes through the duct throttle part **30**, the water separated by the blocking wall **24** and having the relatively small particles is guided to the body-side drainage outlets **34** by the partition wall **26** and is drained out of the cowl duct part **16**. Thus, the dry air flows as the airflow AR**3** from the air outlet **12**H to the air-conditioning duct **44** and is supplied into the vehicle interior. When the water flowing along the partition wall **26** in the duct throttle part **30** is not drained from the body-side drainage outlets **34**, and flows toward the air outlet **12**H across the body-side drainage outlets **34**, the flow of the water toward the air outlet **12**H is hindered by the inclined wall **36**. The water flows backward along the inclined wall **36** toward the body-side drainage outlets **34**, and is drained out of the cowl duct part **16** from the body-side drainage outlets **34**.
(38) As described above, the gas-liquid separation mechanism **20** of the cowl structure **10** includes

the chamber **22** having the interior serving as the chamber interior **28**, and the chamber **22** is provided in the bottom part of the cowl body **12** and is disposed between the outside air inlets **14**C and the air outlet **12**H in the vehicle width direction. The chamber **22** includes the blocking wall **24** constituting a right wall of the chamber **22** and having, at its lower end, the water inlets **24**A, the chamber-side drainage outlets **32**, and the communication ports **26**A that allows the chamber interior **28** to communicate with the inside of the cowl body **12** on the left side of the blocking wall **24**. Consequently, as described above, the airflow AR**1** containing the liquid water having the relatively large particles can be caused to flow into the chamber interior **28** from the water inlets **24**A of the blocking wall **24**, thus achieving separation between water and air. Then, the separated air can be returned from the chamber interior **28** to the cowl duct part **16** through the communication ports **26**A and caused to flow toward the air outlet **12**H, and the separated water can be drained out of the cowl duct part **16** from the chamber-side drainage outlets **32**. Furthermore, the airflow AR**2** containing the water scattered in droplets can be caused to hit the blocking wall **24** of the chamber **22**, thus achieving separation between the air and the water contained in the airflow AR**2**. That is, in the cowl structure **10** of the present embodiment, the water and the air can be separated from each other in the inside of the chamber interior **28** for the airflow AR**1** containing the liquid water having the relatively large particles, and the water and the air can be separated from each other by the blocking wall **24** for the airflow AR**2** containing the water scattered in droplets. As described above, according to the cowl structure **10** of the present embodiment, it is possible to achieve satisfactory separation between the liquid and the air.

(39) In addition, the blocking wall **24** is inclined rightward (outside air inlet **14**C side) toward the upper side when viewed from the front side. Consequently, as described above, it is possible to create a vortex flow directed downward when the airflow AR**2** hits the blocking wall **24**. As a result, the vortex flow allows water droplets and air contained in the airflow AR**2** to be effectively separated from each other.

(40) In addition, the chamber **22** is disposed at a position spaced apart downward from the cowl top panel **14**, and the partition wall **26** of the chamber **22** extends leftward from the upper end of the blocking wall **24**. Furthermore, the partition wall **26** is inclined downward toward the left side when viewed from the front side, and the lower end of the partition wall **26** is coupled to the bottom wall **12**A of the cowl body **12**. Consequently, in the cowl duct part **16**, the duct throttle part **30** can be located above the chamber **22**. Thus, the atomized water separated by the blocking wall **24** and having small particles can be drawn into the duct throttle part **30** by the airflow AR**3** flowing through the duct throttle part **30**. Furthermore, the water drawn into the duct throttle part **30** can be guided to the bottom wall **12**A of the cowl body **12** along the bottom surface of the duct throttle part **30** (the upper surface of the partition wall **26**), and be drained from the body-side drainage outlets **34** of the cowl body **12**.

(41) In addition, the up-down dimension H**1** of the blocking wall **24** of the chamber **22** is set to be larger than the up-down dimension H**2** of the inlet **30**A of the duct throttle part **30**. That is, an area of the blocking wall **24** when viewed in the left-right direction is set to be larger than an area of the inlet **30**A of the duct throttle part **30**. This can increase the flow velocity of the airflow AR**3** flowing through the inlet **30**A of the duct throttle part **30**. Consequently, the atomized water separated by the blocking wall **24** and having the small particles can be satisfactorily drawn into the duct throttle part **30** by the airflow AR**3**.

(42) The inclined wall **36** is provided in the bottom part of the cowl body **12** between the body-side drainage outlets **34** and the air outlet **12**H. The inclined wall **36** is inclined upward toward the air outlet **12**H when viewed from the front side, and is disposed below the air outlet **12**H. This allows the water contained in airflow AR**3** that has passed through the duct throttle part **30** to be returned toward the body-side drainage outlets **34** by the inclined wall **36** and to be drained from the body-side drainage outlets **34** even if the water passes the body-side drainage outlets **34** to the left without being drained from the body-side drainage outlets **34**.

(43) In the present embodiment, although an inclination angle of the blocking wall **24** with respect to the bottom wall **12**A of the cowl body **12** is not particularly defined, for example, the inclination angle of the blocking wall **24** can be appropriately changed according to an up-down dimension of the cowl structure **10**.
(44) In addition, from the viewpoint of satisfactorily creating a vortex flow after the airflow AR**2** hits the blocking wall **24**, thus achieving separation between the air and the liquid in the airflow AR**2**, the blocking wall **24** is desirably disposed to be inclined rightward toward the upper side, but the blocking wall **24** may be disposed along the up-down direction when viewed from the front side. Even in this case, the air and the liquid contained in the airflow AR**2** can be separated from each other by causing the airflow AR**2** to hit the blocking wall **24**.
(45) In the present embodiment, the bottom wall **12**A of the cowl body **12** may be slightly inclined downward toward the left side when viewed from the front side. This can cause liquid water flowing along the bottom wall **12**A to satisfactorily flow into the chamber interior **28** of the chamber **22**.
(46) Furthermore, in the present embodiment, one chamber **22** is applied in the cowl structure **10**, but two chambers **22** may be applied corresponding to various vehicles. For example, as illustrated in FIG. **5**, when the outside air inlets **14**C are at respective vehicle-widthwise ends of the cowl top panel **14**, and the air outlet **12**H is at the vehicle-widthwise center part of the cowl body **12**, each of the chambers **22** may be disposed between each outside air inlet **14**C and the air outlet **12**H. In this case, the chambers **22** are disposed symmetrically with respect to the vehicle-widthwise center part of the cowl structure **10**.
(47) Furthermore, in the present embodiment, the chamber **22** is disposed on the right part of the cowl body **12**, but the chamber **22** can be set at any position. For example, the chamber **22** can be set at any position corresponding to a wiper unit or an airbag apparatus to be mounted in the cowl body **12**.
(48) In the present embodiment, although the chamber **22** and the inclined wall **36** are formed in one piece in the gas-liquid separation mechanism **20**, the chamber **22** and the inclined wall **36** may be formed in separate pieces.

Claims

1. A vehicle cowl structure for a vehicle comprising: a cowl body extending in a vehicle width direction on a vehicle front side of a lower end of a windshield glass of the vehicle and having a recessed shape with an upper opening facing a vehicle upper side when viewed in a longitudinal direction of the cowl body; a cowl panel extending in the vehicle width direction on the vehicle upper side of the cowl body, closing the upper opening of the cowl body, and having an outside air inlet configured to allow outside air to be introduced into the cowl body; an air outlet disposed in the cowl body and configured to allow air inside the cowl body to be discharged toward a vehicle interior of the vehicle; and a chamber having a hollow interior serving as a chamber interior, provided in a bottom part of the cowl body, and disposed between the outside air inlet and the air outlet in the vehicle width direction, wherein the chamber comprises a blocking wall constituting a wall of the chamber on a side of the outside air inlet, extending upward from a bottom wall of the cowl body, and having, at a lower end of the blocking wall, a water inlet configured to allow liquid inside the cowl body to be drawn into the chamber interior, a chamber-side drainage outlet configured to allow the liquid drawn into the chamber interior to be drained out, and a communication port disposed on a side of the air outlet with respect to the blocking wall and configured to allow the chamber interior to communicate with an inside of the cowl body.
2. The vehicle cowl structure according to claim 1, wherein the blocking wall is inclined toward the outside air inlet in a direction toward the vehicle upper side when viewed from the vehicle front side.

3. The vehicle cowl structure according to claim 2, wherein the chamber comprises a partition wall disposed at a position spaced apart from a vehicle lower side of the cowl panel and extending from an upper end of the blocking wall toward the air outlet, and the partition wall is inclined toward the vehicle lower side in a direction toward the air outlet when viewed from the vehicle front side, and has a lower end coupled to the bottom wall of the cowl body.
4. The vehicle cowl structure according to claim 2, wherein the cowl body has, in the bottom wall of the cowl body, a body-side drainage outlet between the chamber and the air outlet, the body-side drainage outlet being configured to allow the liquid inside the cowl body to be drained out.
5. The vehicle cowl structure according to claim 3, wherein the cowl body has, in the bottom wall of the cowl body, a body-side drainage outlet between the chamber and the air outlet, the body-side drainage outlet being configured to allow the liquid inside the cowl body to be drained out.
6. The vehicle cowl structure according to claim 4, wherein the cowl body comprises, in the bottom part of the cowl body, an inclined wall between the body-side drainage outlet and the air outlet, and when viewed from the vehicle front side, the inclined wall is inclined toward the vehicle upper side in a direction toward the air outlet, and is disposed on a vehicle lower side relative to the air outlet.
7. The vehicle cowl structure according to claim 5, wherein the cowl body comprises, in the bottom part of the cowl body, an inclined wall between the body-side drainage outlet and the air outlet, and when viewed from the vehicle front side, the inclined wall is inclined toward the vehicle upper side in a direction toward the air outlet, and is disposed on the vehicle lower side relative to the air outlet.